

# SCIENTIFIC REFEREE REPORT

## The Face-Adjacency Graph of the Truncated Octahedron

Laplacian spectrum, symmetry decomposition, an inter-orbit operator, and the magnetic Laplacian

Luke Martin — Draft 10, October 2026

**Editorial recommendation: CONDITIONAL MINOR REVISION.** No critical or major mathematical error was established in the claims examined. The new bridge from rule-defined matrices to the existing formal matrix theorems is mathematically well supported. The final release cannot be regarded as independently reproduced until the version-pinned executable Python and Lean project has actually been obtained and run.

**Review scope.** Fourteen-page submitted PDF, comparison against Draft 9, inspection of the publicly accessible source of `Graph.lean` at the stated `face-graph-note-v10` tag, independent reconstruction from the geometric polyhedron and the stated gauge, and targeted negative-control checks. The reviewer could not download the complete tagged code release into the execution environment and has *not* run the Lean kernel or the author’s complete Python runner. A previously developed finite-mathematics regression suite was run again, but its previously supported results are not represented as independent proof of the new formal source.

**Important qualification.** The phrase “kernel checked” in the manuscript describes the *author’s reported* formal build, not a kernel run independently witnessed in this review. Conversely, inability to access the full release in this environment is not evidence that its programs fail. No aggregate numerical score is assigned; the findings and criterion assessments are the substance of this recommendation.

## 1 Executive scientific assessment

This paper gives a clear finite example of how a non-regular polyhedral face graph can be decomposed using octahedral symmetry, and how a magnetic connection changes its Laplacian spectrum. Its ordinary graph is the tetrakis-hexahedral graph on 14 vertices and 36 edges; its principal value is not the discovery of that graph but the explicit Laplacian decomposition, the inter-orbit operator  $K$ , and the exceptional exact quarter- and half-flux magnetic spectral factorizations.

Draft 10 makes one significant new contribution to the *verification of the manuscript*: the new `Graph.lean` file defines the adjacency relations and special-flux phase formula as functions of the labeled vertices, derives finite matrices from those functions, and states equality theorems connecting the derived matrices to the previously hand-entered Lean literals. If the pinned Lean build works as reported, this materially reduces the formerly acknowledged transcription gap. The remaining gap—between those combinatorial rules and their realization by the actual polyhedron, including oriented plaquette holonomies—is accurately left outside the claimed Lean scope.

The independent review tested the bridge along a different route: it reconstructed the faces directly as subsets of the 24 vertices  $(0, \pm 1, \pm 2)$  and required every geometric face intersection to produce the same adjacency entries as the rules transcribed in `Graph.lean`. It reconstructed the quarter-flux matrix twice, checked all 24 physical oriented triangle holonomies, and confirmed the three graph-defined characteristic polynomials. These tests passed and deliberately mutated constructions failed, as discussed below. This is considerably more informative for Draft 10 than another stand-alone calculation of the ordinary polynomial.

## 2 Contribution and changed-material assessment

### 2.1 What changed from Draft 9

A textual comparison of the supplied Draft 9 and Draft 10 PDFs shows the central statements, proof calculations, magnetic formulas, and spectral plot are substantively unchanged. The new material concerns `Graph.lean`, its claims of definitional matrix equivalence, the expanded proof-status disclosures in Section 8, Table 3 on Lean theorem-to-claim coverage, and a corresponding refinement of acknowledgments. The targeted review therefore concentrates on these additions and on their dependency on the existing mathematics.

### 2.2 Ordinary graph and spectral blocks

Label the square faces by  $(a, \sigma)$  with  $a \in \{1, 2, 3\}$  and  $\sigma = \pm 1$ , and the hexagonal faces by sign vectors  $h \in \{\pm 1\}^3$ . A square is adjacent to  $h$  exactly when  $h_a = \sigma$ , and hexagons are adjacent precisely when they differ in one sign. This gives 24 square–hexagon and 12 hexagon–hexagon edges. As the paper establishes, the resulting  $O_h$  representation decomposes as

$$\mathbb{R}^{14} \cong 2A_{1g} \oplus E_g \oplus 2T_{1u} \oplus T_{2g} \oplus A_{2u}.$$

On each of the three  $T_{1u}$  multiplicity spaces, the ordinary Laplacian is unitarily equivalent to

$$L_{T_{1u}} = \begin{pmatrix} 4 & -2 \\ -2 & 5 \end{pmatrix}, \quad \det(xI - L_{T_{1u}}) = x^2 - 9x + 16.$$

The remaining components contribute  $x(x-7)$ ,  $(x-4)^2$ ,  $(x-7)^3$ , and  $(x-9)$ , yielding

$$\det(xI - \Delta) = x(x-9)(x-7)^4(x-4)^2(x^2 - 9x + 16)^3. \quad (1)$$

This decomposition, its multiplicities and the splitting field  $\mathbb{Q}(\sqrt{17})$  are consistent with the computations examined across prior drafts. The new audit independently confirmed that the rule-defined  $14 \times 14$  Laplacian has exactly polynomial (1); the additional advance in Draft 10 is the proposed machine-checkable bridge from these rules to the literate formal matrices.

## 3 Focused examination of `Graph.lean`

### 3.1 The three distinct mathematical layers

It is particularly important to keep three assertions apart:

- (a) **Geometry to rules.** The faces of the truncated octahedron are represented by the chosen labels, and two faces share an edge precisely under the stated adjacency predicates. This is Proposition 2.1 and is not formalized in the new Lean file.
- (b) **Rules to matrices.** The predicates `sqHexAdj`, `hexHexAdj` and `adjN` determine `Agraph`, while the fixed degrees give `Lgraph` and `Sgraph`. The real and imaginary parts of the quarter-flux construction determine `Dre`, `Dim` and hence  $\Delta_6$ . `Graph.lean` states equality theorems identifying these with `Lfin`, `Sfin` and `D6`.
- (c) **Matrices to characteristic polynomials.** The previously introduced Lean files assert the exact polynomials for the literal matrices. The new file rewrites the rule-defined matrices by the equality theorems and applies these existing characteristic-polynomial theorems.

The distinction matters because proof of (b) and (c) does not automatically constitute a fully machine-checked proof of (a). Draft 10 makes precisely this disclosure in Section 8 (pp. 11–12). It should be retained in any final version.

### 3.2 Independent geometric test of the first layer

Starting solely from the paper’s coordinate model, construct the 24 distinct vertices as all permutations of  $(0, \pm 1, \pm 2)$ . Each of the six square face equations  $x_a = 2\sigma$  selects four vertices; each of the eight hexagon planes  $h \cdot x = 3$  selects six. Declare two faces adjacent if and only if their sets of geometric vertices intersect in exactly two points. This recovers the adjacency matrix generated by the Lean rule `adjN entry by entry`. In particular, this test is sensitive to the indexing of the three sign bits, rather than merely to the unlabeled graph isomorphism class.

The bit convention in the public source is

$$h_j = \begin{cases} +1, & \lfloor h/2^{2-j} \rfloor \bmod 2 = 0, \\ -1, & \text{otherwise,} \end{cases} \quad \sigma(s) = \begin{cases} +1, & s \text{ even,} \\ -1, & s \text{ odd.} \end{cases}$$

For the finite labeled system, the resulting degree vector is  $(4^6, 6^8)$  and the adjacency matrix is symmetric. An intentionally reversed bit-order convention breaks the *labeled* geometric adjacency comparison, providing a useful negative control for the indexing assumption.

### 3.3 Independent magnetic test of the new quarter-flux bridge

For a square  $(a, \sigma)$  adjacent to a hexagon  $h$ , with  $(a, b, c)$  cyclic, the manuscript’s incidence phase is

$$B_{(a,\sigma),h} = \frac{h_b - i\sigma h_c}{1 - i\sigma} = \frac{h_b + h_c}{2} + i\sigma \frac{h_b - h_c}{2}. \quad (2)$$

The source defines the real and imaginary parts on the right using integer arithmetic. On every admissible edge the numerators are  $-2, 0$ , or  $2$ , so the divisions by  $2$  are exact; the resulting values are  $\pm 1$  or  $\pm i$ . Its block matrix is

$$\Delta_6 = \begin{pmatrix} 4I_6 & -B \\ -B^* & 6I_8 - C \end{pmatrix},$$

where  $C$  is the cube adjacency matrix. The independent test obtained the same matrix from the rule-based real and imaginary parts, and established

$$B^*B = \frac{9I_8 - C^2}{2}, \quad BB^* = 4I_6.$$

In addition, each of the 24 triangles was oriented outward using the determinant of its three face-centre vectors. For an outward triangle  $(s, h, h')$ , the product  $B_{s,h}\overline{B_{s,h'}}$  is exactly  $i$ . Hence the matrix constructed from the formula really represents the  $q = 6$  plaquette holonomy claimed in Section 6.1, *as an independent exact check*. A sign flip in just one nonzero  $B$  entry destroys both a Gram identity and at least one triangular holonomy, establishing that these tests detect relevant mutations.

The corresponding block reduction gives three copies each of

$$\begin{pmatrix} 4 & -2 \\ -2 & 5 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 4 & -2 \\ -2 & 7 \end{pmatrix}$$

and two remaining eigenvalues  $3$  and  $9$ . This yields

$$\det(xI - \Delta_6) = (x - 9)(x - 8)^3(x - 3)^4(x^2 - 9x + 16)^3. \quad (3)$$

At  $q = 12$ , the all-negative phase gauge produces  $D + A$  and therefore

$$\det(xI - \Delta_{12}) = (x - 8)^3(x - 5)^3(x - 4)^2(x - 3)^4(x^2 - 13x + 24). \quad (4)$$

Both polynomials were independently recomputed using exact finite matrices. No new contradiction in the three characteristic polynomials was found.

## 4 Topological argument and dependency boundaries

Proposition 6.1 is explicitly about a connected graph *cellularly embedded in the sphere*. For the physical face graph,  $V = 14$ ,  $E = 36$ , and the 24 outward-oriented triangle boundaries define boundary maps  $\partial_2 : \mathbb{Z}^{24} \rightarrow \mathbb{Z}^{36}$  and  $\partial_1 : \mathbb{Z}^{36} \rightarrow \mathbb{Z}^{14}$ . Earlier exact review evidence found

$$\partial_1 \partial_2 = 0, \quad \text{rank } \partial_1 = 13, \quad \text{rank } \partial_2 = 23 = 36 - 14 + 1.$$

Thus the face boundaries span the cycle space, which supports gauge uniqueness for prescribed spherical face holonomies. The manuscript correctly includes the non-spherical toroidal winding-holonomy control, showing why the hypothesis cannot be discarded. These findings are inherited and were selectively rerun, not rediscovered as a new result of Draft 10.

The additional Lean bridge does not formalize spherical cell attachment, the outward face boundaries, the global gauge classification, or Proposition 6.2's third-moment classification. Nor does it formalize all representation-theoretic labels or the full eigenspace-intertwining claim for  $K$ . These are not mathematical errors: the manuscript's current Table 2 and proof-status paragraph disclose the boundaries. An editor should assess the work as a combination of self-contained hand arguments, exact computer calculations and a *partially* formalized finite-matrix core.

## 5 Evidence status and reproducibility limitation

| Evidence item                                                      | Observed status        | What this does and does not establish                                                                                       |
|--------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Original Draft 10 PDF and Draft 9 comparison                       | Inspected and compared | Identifies the new claims and protects previously checked statements from repeated, inflated credit.                        |
| Public <code>Graph.lean</code> at <code>face-graph-note-v10</code> | Source inspected       | New adjacency, gauge and equality theorem statements are present; successful compilation is not established.                |
| Independent graph-to-geometry bridge tests                         | 44/44 passed           | Confirms the submitted finite formulas by independent construction, not Lean proof checking.                                |
| Inherited targeted finite-mathematics controls                     | 25/25 passed           | Corroborates unchanged spectra, $K$ and topology; some checks duplicate the new test for deliberate regression coverage.    |
| Full author Python runner                                          | Not executed           | Its stated integration and complete dependencies remain to be tested in a clean checkout.                                   |
| Pinned Lean/Mathlib build                                          | Not executed           | No independent assertion can be made that <code>lake build</code> completed or that the reported axiom lists were obtained. |
| Zenodo deposit for Draft 10                                        | Not established        | Manuscript states identifier will be supplied when the deposit exists.                                                      |

The principal blocker is environmental access: the reviewing execution environment could inspect public raw source via a web retrieval channel, but could not download the complete tagged project using its local network. The independent scientific checks and this report should not be described as evidence of a failed author build.

## 6 Fourteen-criterion scientific assessment

The criteria below are *individual editorial judgments*, not a statistical calibration and not a probability of manuscript correctness.

| Criterion                        | Assessment            | Reason                                                                                                                           |
|----------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Mathematical correctness         | Strong on scope       | No contradiction found in the examined finite claims; graph and flux matrix identities survive independent exact checks.         |
| Proof completeness               | Good, scoped          | Printed graph, block and gauge arguments are detailed; not all propositions are formalized, and this is disclosed.               |
| Definitions and hypotheses       | Strong                | Spherical cellular embedding, vertex labels, flux sector and allowable gauges are explicit.                                      |
| Originality                      | Promising, bounded    | Known graph and monopole method distinguished from the specific exact factorizations; targeted prior-art work is not exhaustive. |
| Significance                     | Focused               | A useful nonregular-graph example and explicit special-flux intertwiner; no broad universality is asserted.                      |
| Exposition and pedagogy          | Strong                | Two-orbit and cube-sector arguments convey why the factors appear, rather than only displaying polynomials.                      |
| Reproducibility                  | Conditional           | Source names, tag, versions and tolerances are stated, but full tagged execution remains unverified here.                        |
| Computational validity           | Good on tested items  | Exact matrix reconstruction and perturbation controls passed; complete upstream scripts not rerun.                               |
| Formal verification              | Improved, conditional | New rule-to-literal bridge is visible and coherent; pinned kernel replay and axiom output remain outstanding.                    |
| Robustness and negative controls | Good                  | Wrong bit order and phase-sign changes are detected; spherical/topological limitation is explicit.                               |
| Literature positioning           | Good, bounded         | Known graph and Avishai–Luck flux work identified; publication priority cannot be proven by finite search.                       |
| Scope discipline                 | Strong                | All-charge spectra are numerical; only selected flux sectors are certified exactly.                                              |
| Figures and tables               | Good                  | Figure 1 joins sorted discrete levels and is not misrepresented as analytical spectral branches.                                 |
| Submission readiness             | Minor release gate    | Complete pinned replay and fixed deposition record remain the two actionable finishing tasks.                                    |

## 7 Actionable findings and dependency assessment

**R1 — Tagged release replay (required).** The paper specifies `face-graph-note-v10`, a Python one-command runner, and Lean 4/Mathlib versions. Obtain a frozen checkout and run, in an uncontaminated environment, installation of the documented dependencies, the documented one-command

Python runner (`run_face_graph_checks.py`), `lake exe cache get` and `lake build` inside the declared Lean project directory. Preserve process exit codes, all named check results, Lean axiom listings and file hashes. A reproducible mismatch in the graph, gauge or characteristic-polynomial data would reopen the corresponding theorems; an environment-only dependency failure would require a corrected recipe but would not alone falsify the mathematics. *This review did not execute these commands against the author release.*

**R2 — Permanent public archival identifier (minor, required for final deposit).** The manuscript still promises a Zenodo archive but leaves its identifier for later insertion. Supply the immutable deposit link/DOI for the exact submitted source, including the Lean tree, scripts, supplementary checkers and plotted data. Record the tag/commit identifier in the deposit. This is a scholarly reproducibility requirement, not a defect in the mathematical argument.

**R3 — Keep the formal-proof boundary explicit (important preservation request).** The new Table 3 is useful because it names `Lgraph_eq_Lfin`, `Sgraph_eq_Sfin`, `Dgraph_eq_D6` and the corresponding `charpoly` theorems, distinguishing them from geometric facts and unformalized representation theory. Retain that distinction; do not generalize the partial finite-matrix certification to a complete kernel proof of the spherical embedding, magnetic flux interpretation or  $O_h$  representation labels.

**R4 — Attribution and presentation (optional).** The attribution paragraphs are unusually detailed for a short mathematical note. Their essential scholarly content—who contributed formulas, that automated-assisted rounds were not independent referees, and that the author retains responsibility—is worth preserving. A journal may prefer to consolidate mechanics and round chronology into supporting correspondence while retaining accurate intellectual credit. This is not an argument for removing relevant attribution.

The logical dependencies are limited. R1 and R2 affect independent verifiability and final release clearance. A failure of `Lgraph_eq_Lfin` would undermine the formal bridge to (2), but not necessarily the printed orbit-block hand proof. Failure of `Dgraph_eq_D6` on correctly defined gauge entries, or a genuine contradiction in the oriented holonomies, would be more serious for the quarter-flux physical interpretation. A Lean compilation issue alone may be fixable without changing a mathematical theorem. R3 is a scope-preservation request, not a newly detected error.

## 8 Editorial conclusion and reversal conditions

**Recommendation: conditional minor revision.** The main mathematics has survived scrutiny over repeated drafts. Draft 10 directly addresses the remaining structural concern about formally proving the same matrices that are defined by the manuscript rather than merely typed as literals. The independent geometry-to-graph and rule-to-magnetic tests here provide additional, adversarially controlled support.

**What would change the recommendation?** A verified counterexample to the geometry-to-rule identification, a valid violation of the oriented quarter-flux holonomies, or a reproducible disagreement between the rule-constructed and hand-entered matrices would trigger major revision and potentially rejection if a central theorem ceased to hold. Discovery of exact prior publication of the special-flux factors would primarily affect originality and journal suitability. Continued inability to reproduce the frozen release would prevent removal of the release condition; it would not, without a demonstrated defect, justify calling the graph theory false.

**What would close the file?** An independently logged, passing complete tagged Python plus Lean build, consistent axiom reports, and the immutable deposit citation, with any necessary version-specific instructions. No additional generalization, extra eigenspectrum calculation, or expanded formalization is required simply to close this round.

## Sources inspected

- [1] L. Martin, *The Face-Adjacency Graph of the Truncated Octahedron*, Draft 10 (submitted PDF), Sections 2–8, Tables 1–3; Draft 9 used for change comparison.
- [2] Public tagged source `Graph.lean`, [GitHub: Graph.lean \(tag v10\)](#); source inspected, *not* compiled in this review.
- [3] E. W. Weisstein, *Tetrakis Hexahedral Graph*, Wolfram MathWorld, <https://mathworld.wolfram.com/TetrakisHexahedralGraph.html>.
- [4] Y. Avishai and J. M. Luck, *Tight-binding electronic spectra on graphs with spherical topology I: The effect of a magnetic charge*, *J. Stat. Mech.* (2008) P06007, DOI [10.1088/1742-5468/2008/06/P06007](https://doi.org/10.1088/1742-5468/2008/06/P06007).